Information Sheet

Machine Learning (Springer)

A Local RAG Retrieval Benchmarking Pipeline for Corpus-Specific Embedding-Model Selection

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1. What is the main claim of the paper? Why is this an important contribution to the machine learning / data mining research area?

The main claim is that embedding-model selection for retrieval-augmented generation (RAG) should be decided per corpus, and that this decision can be made repeatable and statistically interpretable by an end-to-end local benchmarking pipeline. The pipeline turns a folder of documents into a complete evaluation benchmark—chunks, synthetic queries, LLM-constructed relevance judgments, multi-model retrieval runs, uncertainty estimates, and automated reports—without requiring the practitioner to supply any labeled data.

This matters because standard practice is to pick an embedding model from a public leaderboard. Public benchmarks are essential shared references, but they measure behavior on public tasks, not on the practitioner's own documents, and they cannot say whether an observed ranking gap on a local corpus is large enough to act on. The contribution is a reusable measurement instrument rather than a new model or metric.

2. What is the evidence provided to support claims? Be precise.

The pipeline is demonstrated through one frozen experimental snapshot, reported in full and released as artifacts:

- **Scope.** Three corpora (`education_eng`, `education_tr`, `law_tr`); 60 source documents; 1,914 chunks; five sentence-transformer candidates (M1–M5); 150 generated synthetic queries; 9,296 judged query–chunk pairs, of which 952 are positive; 148 positive-bearing queries scored after zero-positive exclusion.
- **Relevance judgments.** Model-independent pooled qrels built from the union of top-20 candidates retrieved by all five models, with explicit negative labels retained so the judged set is not biased toward any one candidate.
- **Uncertainty.** Percentile and paired bootstrap intervals ($B = 10{,}000$) over each corpus's own scored queries. The paired intervals are the discriminating evidence: on `education_eng` the M1−M2 interval includes zero, so that corpus does not separate the two leading candidates on retrieval quality and the decision falls to latency; on both Turkish corpora every paired interval against M5 excludes zero.
- **Pool completeness.** Seed-chunk coverage is measured directly rather than assumed: 9 of 150 seed chunks fall outside the pooled candidate set.
- **Operational cost.** Compound corpus embedding and index-insertion throughput, and compound query-embedding and vector-search latency, from a single-run CPU-only snapshot.
- **Verification.** Every hand-written number in the manuscript is checked against the stored run artifacts by an automated gate (33 values); the reported snapshot is frozen and released.

The paper is explicit about what this evidence does not establish: queries are synthetic and qrels are LLM-generated without human validation; each language and domain condition is represented by one corpus, so both are confounded with corpus identity; retrieval geometry, chunking, pool depth, and cutoff are fixed demonstration settings with no sensitivity study; and timing is single-run and single-machine. These limits appear in the abstract and Section 6.4.

3. Report the 3–5 most closely related contributions in the past 7 years, and briefly state the relation of the submission to them.

1. **Muennighoff et al. (2023), MTEB**, EACL; and **Enevoldsen et al. (2025), MMTEB**, ICLR. These supply corpus, queries, and judgments together and answer which model performs best on fixed public tasks. This submission is upstream and complementary: it *produces* a labeled benchmark from a practitioner's own unlabeled documents—precisely the input MTEB-style evaluation presupposes.
2. **Thakur et al. (2025), Assessing Support for the TREC 2024 RAG Track**, SIGIR. Characterizes when LLM relevance assessment agrees with human assessment. This submission depends on LLM judging and cites this work as the basis for treating its own unvalidated qrels as a stated limitation rather than as ground truth.
3. **Upadhyay et al. (2024), UMBRELA**. Provides an LLM relevance judge as a component. This submission uses a judge of that kind inside a larger corpus-to-report workflow and contributes the surrounding pooling, uncertainty, and reporting stages.

4. **Es et al. (2024), RAGAs**, EACL Demos. Scores an assembled generation pipeline on faithfulness and answer relevance, and can synthesize evaluation data from documents. This submission sits upstream: it compares ranked retrieval over the full chunk set, the stage RAGAS assumes is already fixed.
5. **Wang et al. (2025), On the Reproducibility Limitations of RAG Systems**. Documents how RAG evaluation results fail to reproduce. This submission responds by freezing and releasing every intermediate artifact and verifying all reported numbers against them automatically.

4. Specify 5 general keywords and 5 specific keywords describing the main research activity.

General: information retrieval; retrieval-augmented generation; evaluation methodology; benchmarking; reproducibility.

Specific: corpus-specific embedding-model selection; synthetic query generation; LLM-as-judge relevance construction; model-independent pooled qrels; query-level paired bootstrap intervals.

5. Conflicts of interest

Institutional conflicts. The author is affiliated with the Department of Software Engineering, TED University, Ankara, Türkiye. Reviewers from TED University should not be assigned.

Guest Editorial Board conflicts. None known.

Financial and personal conflicts. None. This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The author declares no competing financial interests or personal relationships that could have appeared to influence the work.

Prior publication. A preliminary version was presented at the International Conference on Computer Science and Intelligence Systems (ICCSIS 2025), Madrid, Spain, 10–11 August 2025, pp. 1–6; the conference organizers subsequently published accepted papers in the *International Journal of Advances in Electronics and Computer Science* 12(9), ISSN 2394-2835. That version established feasibility on a single English corpus. The present submission extends it with three language- and domain-differentiated corpora, model-independent pooled qrels, query-level bootstrap uncertainty, dataset-level reports as the primary selection artifact, and operational-cost evidence. The relationship is stated in Section 1 and in full in Section 2.6, and the prior paper is cited and supplied with this submission. The manuscript is not under review at any other journal or conference.

6. Suggested reviewers (up to four)

TO BE COMPLETED BY THE AUTHOR BEFORE UPLOAD. Machine Learning requires up to four candidate reviewers. Supply full name, affiliation, email, and a one-line justification for each. None may be from TED University, a co-author, or a recent collaborator.

1. Name — Affiliation — Email — Why appropriate:

2. Name — Affiliation — Email — Why appropriate:

3. Name — Affiliation — Email — Why appropriate:

4. Name — Affiliation — Email — Why appropriate: